ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 3 SET B **WINTER SEMESTER, 2024-2025**

DURATION: 30 MINUTES **FULL MARKS: 30**

MATH 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

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1. The course teacher of Math 4341 of SWE'23 has recently observed an unusual pattern in student attendance. He has noticed that many students are frequently absent from class and is concerned that, if this trend continues, some students may not meet the attendance requirements to sit for the semester final examination. With the final exam about one month away, the instructor aims to predict students' future attendance in order to identify and warn those who may be at risk.

   According to information provided by anonymous but reliable Linear Algebra enthusiasts, the **number of classes attended (r)** is influenced by the **number of hours a student spends watching reels or short-form videos (p)** and **the number of Gilbert Strang lectures the student has watched (q)**. To investigate this relationship, the following data (see Table 1) have been provided as supporting evidence, and the instructor considers this information credible.

Table 1: Data for Attendance Prediction

Student	Reel Watch Time (p)	Class Lectures Watched (q)	Classes Attended (r)
Samiul	10	7	11
Asif	9	4	0
Emon	12	8	15
Jarif	6	3	7

Let the equation for predicting attendance be of the form $r = mp + nq + s$, where m, n, and s are constants to be determined. Now answer the following questions:

a) Write down the data from Table 1 in Matrix form $Ax = b$, where A is the coefficient matrix, x is the variable vector, and b is the constant vector. 1 (CO3) (PO1)

Solution:

The data can be represented in matrix form as follows (some rows were rearranged which won't affect the solution):

$$\underbrace{\begin{bmatrix} 10 & 7 & 1 \\ 9 & 4 & 1 \\ 12 & 8 & 1 \\ 6 & 3 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} m \\ n \\ s \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 11 \\ 0 \\ 15 \\ 7 \end{bmatrix}}_{b}$$

Since,

$$\begin{bmatrix} p & q & 1 \end{bmatrix} \begin{bmatrix} m \\ n \\ s \end{bmatrix} = mp + nq + s = r$$

Rubrics:

- 1 marks for correctly writing down the given data in $Ax = b$ format.

b) Based on the data provided, determine the values of m, n, and s using the least squares regression method by solving $A\hat{x} = \hat{b}$ for $\hat{x}$. Note that original vector b may not be in the column space of A, so you need to find $\hat{x}$ for the best approximation of b, $\hat{b}$, in the column space of A. 2 (CO3) (PO1)

Solution:

$$\hat{x} = (A^T A)^{-1} A^T b$$

$$\Rightarrow \hat{x} = \begin{bmatrix} -295/93 \\ 162/31 \\ 823/93 \end{bmatrix} = \begin{bmatrix} -3.17204301 \\ 5.22580645 \\ 8.84946237 \end{bmatrix}$$

Rubrics:

- 2 marks for correctly computing $\hat{x}$ using the least squares method.

c) Find out the best approximation of b, $\hat{b}$ or p, in the column space of A using the value of $\hat{x}$ obtained in the previous part. (p is the projection of b onto the column space of A ($C(A)$)) 1 (CO3) (PO1)

Solution:

$$p = A\hat{x}$$

$$\Rightarrow p = \begin{bmatrix} 425/31 \\ 112/93 \\ 1171/93 \\ 511/93 \end{bmatrix} = \begin{bmatrix} 13.70967742 \\ 1.20430108 \\ 12.59139785 \\ 5.49462366 \end{bmatrix}$$

Rubrics:

- 1 marks for correctly computing p.

d) Find out the matrix P that projects b into the column space of A ($C(A)$). 1.5 (CO3) (PO1)

Solution:

$$P = A(A^T A)^{-1} A^T$$

$$\Rightarrow P = \begin{bmatrix} 0.565 & -0.194 & 0.387 & 0.242 \\ -0.194 & 0.914 & 0.172 & 0.108 \\ 0.387 & 0.172 & 0.656 & -0.215 \\ 0.242 & 0.108 & -0.215 & 0.866 \end{bmatrix} = \begin{bmatrix} 35/62 & -6/31 & 12/31 & 15/62 \\ -6/31 & 85/93 & 16/93 & 10/93 \\ 12/31 & 16/93 & 61/93 & -20/93 \\ 15/62 & 10/93 & -20/93 & 161/186 \end{bmatrix}$$

Rubrics:

- 1 marks for correctly computing P.

e) Write down the expressions for determining the least squares solution $\hat{x}$, projected vector p 1.5 (CO3) (PO1)

and the projection matrix P. (If you have already mentioned these in Question 1.b, Question 1.c and Question 1.d, just highlight those expressions.)

Solution:

Expressions:

- Least squares solution: $\hat{x} = (A^T A)^{-1} A^T b$
- Projected vector: $p = A\hat{x}$
- Projection matrix: $P = A(A^T A)^{-1} A^T$

Rubrics:

- 0.5 marks (times 3) for each correct expression.

f) Find the error in your prediction, i.e. the difference in attendance between the actual data and the predicted data. 2 (CO3) (PO1)

Solution:

$$\text{Error}, e = b - p$$

$$\Rightarrow \; e = \begin{bmatrix} 11 \\ 0 \\ 15 \\ 7 \end{bmatrix} - \begin{bmatrix} 425/31 \\ 112/93 \\ 1171/93 \\ 511/93 \end{bmatrix}$$

$$\Rightarrow \; e = \begin{bmatrix} -84/31 \\ -112/93 \\ 224/93 \\ 140/93 \end{bmatrix} = \begin{bmatrix} -2.70967742 \\ -1.20430108 \\ 2.40860215 \\ 1.50537634 \end{bmatrix}$$

Rubrics:

- 2 marks for correctly computing e.

g) Find the Mean Square Error(MSE) $= \frac{1}{n} \sum_{i=1}^{n} (b_i - \hat{b}_i)^2$. 1 (CO3) (PO1)

Solution:

$$e = \begin{bmatrix} -84/31 \\ -112/93 \\ 224/93 \\ 140/93 \end{bmatrix}$$

$$\therefore MSE = \frac{1}{4}\left(\left(\frac{-84}{31}\right)^2 + \left(\frac{-112}{93}\right)^2 + \left(\frac{224}{93}\right)^2 + \left(\frac{140}{93}\right)^2\right)$$

$$\Rightarrow \; MSE = \frac{392}{93} = 4.215053763$$

Rubrics:

- 1 marks for correctly computing MSE.

2. If a student needs to attend **at least 26 classes** to be eligible to sit for the semester final examination, determine which of the following students are at risk of not meeting the attendance requirements based on your predictions from the answer obtained in question Question 1. 3 (CO2) (PO1)

Table 2: Attendance Data for Risk Assessment

Student	Reel Watch Time (p)	Class Lectures Watched (q)	Classes Attended (r)	Eligibility Status (Yes/No)
Muhaiminul	3	5	**25.46236559**	No
Dhrubo	10	0	**-22.87096774**	No
Wasi	5	7	**29.56989247**	Yes

Solution:

To determine the predicted attendance for the given students and their eligibility status we can replace the values of p and q in the equation $r = mp + nq + s$ using the values of m, n, and s obtained from the least squares regression in Question 1.b. Or simply perform $B\hat{x}$ where B is the matrix of p and q values for the new students.

$$B = \begin{bmatrix} 3 & 5 & 1 \\ 10 & 0 & 1 \\ 5 & 7 & 1 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} -295/93 \\ 162/31 \\ 823/93 \end{bmatrix}$$

$$\Rightarrow B\hat{x} = \begin{bmatrix} 25.46236559 \\ -22.87096774 \\ 29.56989247 \end{bmatrix}$$

Based on the predicted attendance, the eligibility status of each student is shown in the Table 2.

Rubrics:

- 0.75 marks (times 3) for correctly computing the predicted attendance of each student.
- 0.25 marks (times 3) for correctly determining the eligibility status of each student.

3. Let X and Y be two subspaces such that $X \subset Y$ in $\mathbb{R}^n$. Let $X^\perp$ and $Y^\perp$ be the orthogonal complements of X and Y respectively in $\mathbb{R}^n$. Then is $X^\perp \subset Y^\perp$? If yes, then justify with proper reasoning supported by example, and if not, what is the correct relationship between $X^\perp$ and $Y^\perp$? 5 (CO2) (PO1)

Solution:

Lets consider a simple example
Let Y be the subspace of $\mathbb{R}^{10}$ occupying a 3 dimensional space and X be the subspace of $\mathbb{R}^{10}$ occupying a 2 dimensional space such that $X \subset Y$. Then, the orthogonal complement of Y, $Y^\perp$, will be a 7 dimensional subspace of $\mathbb{R}^{10}$, and the orthogonal complement of X, $X^\perp$, will be an 8 dimensional subspace of $\mathbb{R}^{10}$. Therefore, we can conclude that $Y^\perp \subset X^\perp$ and not the other way around since $Dim(X^\perp) > Dim(Y^\perp)$.
So, $X^\perp \not\subset Y^\perp$ rather $Y^\perp \subset X^\perp$.

Rubrics:

- 2 marks correctly commenting on the question whether $X^{\perp} \subset Y^{\perp}$.
- 3 marks for correctly identifying and explaining the correct relationship $Y^{\perp} \subset X^{\perp}$.

4. Let $\vec{a}$ and $\vec{b}$ be two vectors. If $\vec{a}$ is stretched by a factor of 5 and $\vec{b}$ is shrinked by a factor of 2 (halved), then what will be its effect on particular fraction $\hat{x}$ and the Projection Matrix P. 2 (CO2) (PO1)

Solution:

Given, $a = 5a$ and $b = \frac{b}{2}$.

Now,

$$\text{new } \hat{x} = \frac{(5a^T)(b/2)}{(5a^T)(5a)}$$

$$\text{new } \hat{x} = \frac{1}{2}\frac{(a^T)(b)}{(a^T)(a)}$$

$$\text{new } \hat{x} = \frac{1}{10}\text{old } \hat{x}$$

Again,

$$\text{new } P = \frac{(5a^T)(5a)}{(5a^T)(5a)}$$

$$\text{new } P = \frac{(a^T)(a)}{(a^T)(a)}$$

$$\text{new } P = \text{old } P$$

Rubrics:

- 1 mark for showing the effect on $\hat{x}$.
- 1 mark for showing the effect on P.
- **If you projected a on b instead of b on a, then you will get the opposite effect, i.e., $\hat{x}$ will be multiplied by 10 and P will remain unchanged. Thats fine as well.**

5. For the provided data in Table 1, do you think instead of $r = mp + nq + s$, a better model for predicting attendance would be $r = \frac{mp}{3} + nq^3 - 5s^2$? Justify your answer mathematically. 7 (CO3) (PO1)
BONUS: 7

Solution:

The new coefficient matrix A should be constructed as follows:

$$A = \begin{bmatrix} \frac{p}{3} & q^3 & -5 \end{bmatrix} \begin{bmatrix} m \\ n \\ s^2 \end{bmatrix}$$

$$A = \begin{bmatrix} 10/3 & 7^3 & -5 \\ 9/3 & 4^3 & -5 \\ 12/3 & 8^3 & -5 \\ 6/3 & 3^3 & -5 \end{bmatrix}$$

b will be unchanged.
So,

$$\hat{x} = (A^T A)^{-1} A^T b = \begin{bmatrix} --3376951/385591 \\ 20122/1385591 \\ -5300723/1156773 \end{bmatrix} = \begin{bmatrix} -8.75785742 \\ 0.05218483 \\ -4.58233638 \end{bmatrix}$$

$$\text{Residual} = b - A\hat{x} = \begin{bmatrix} -238380/385591 \\ 25520/1156773 \\ 464000/1156773 \\ 225620/1156773 \end{bmatrix} = \begin{bmatrix} -0.61821982 \\ 0.02206137 \\ 0.40111586 \\ 0.19504259 \end{bmatrix}$$

$$Therefore \text{NEW MSE} = \frac{168200}{1156773} = 0.145404$$

Since the MSE of the new model is less than the MSE of the old model, we can conclude that the new model is better for predicting attendance.

Rubrics:

- 1 mark for correctly determining new A.
- 2 marks for correctly calculating new $\hat{x}$.
- 3 marks for correctly calculating anything that leads to error e and determining the new MSE.
- 1 mark for correctly comparing the MSE of the new model with the old model and concluding which one is better.

6. In a 23 dimensional vector space ($\mathbb{R}^{23}$), what is the maximum number of mutually orthogonal 4 dimensional subspaces in the given vector space? 3 (CO2) (PO1)
BONUS: 3

Solution:

The maximum number of mutually orthogonal 4 dimensional subspaces in the given vector space is 5. ($floor(23/4)$)

Rubrics:

- 1 mark for correct answer.
- 2 mark for correct solution approach.

Please note that -

- **Note that: Question 1-4 are mandatory worth 20 marks and Question 5 and 6 are bonus worth additional 10 marks.**
- **Try keeping fractions instead of converting to decimals in Questions 1, 2, and 5 (except for the Projection Matrix) for more accurate results.**
- **The relationships between the names mentioned in the questions and the associated data are 'mostly' hypothetical.**